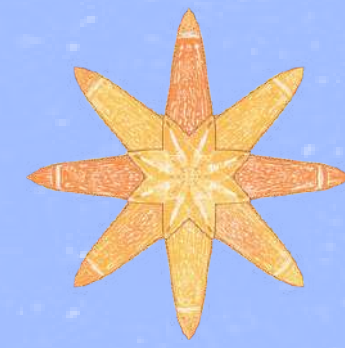
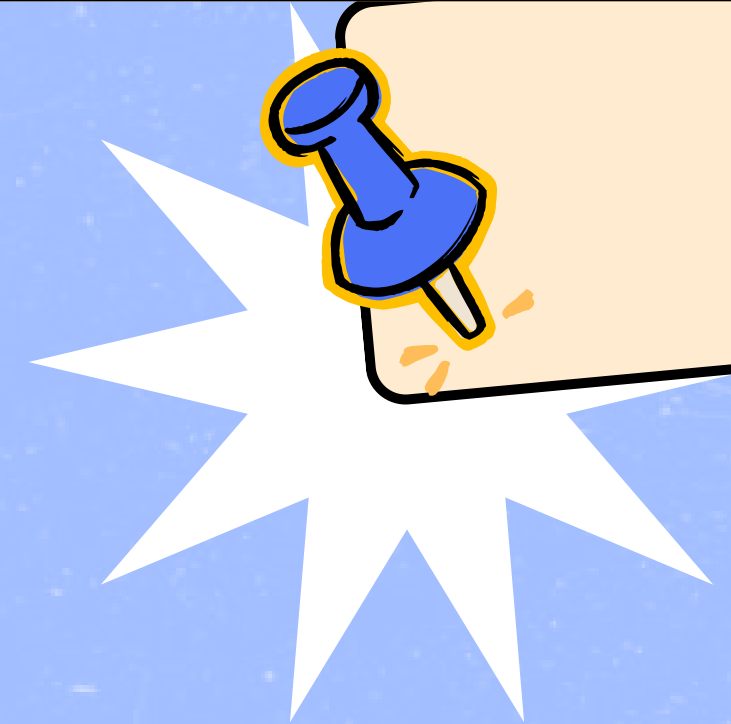




BINARY NUMBERS



COUNT





WHAT ARE NUMBERS?

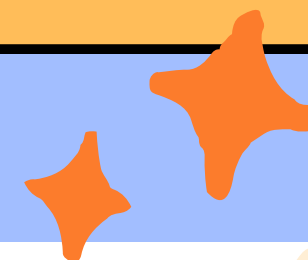


Numbers Are Everywhere Around Us!

Think about counting your fingers, steps you take, or toys you have. Numbers help us count, measure, and describe things in our world!

We Use Numbers Every Single Day

Numbers help us understand and describe the world around us. From counting friends to measuring distances, numbers are our helpful tools!





Binary Numbers



DIFFERENT WAYS TO COUNT



Decimal System

We use base 10 every day with digits 0-9. Count your fingers — that's decimal!

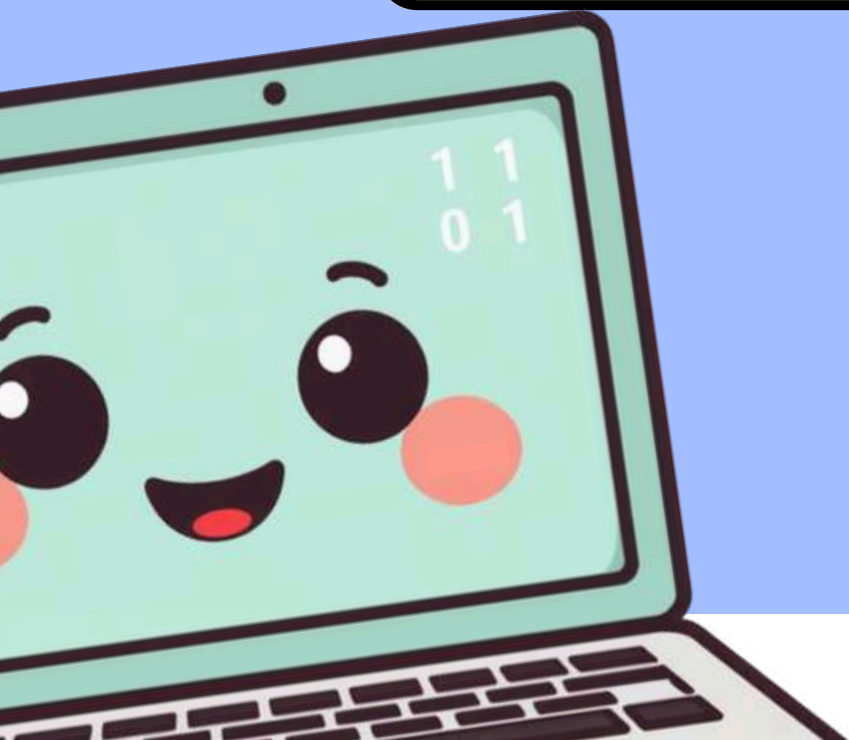
Binary System

Computers use base 2 with only 0 and 1. Simple but powerful!

Side by Side

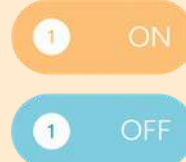
Decimal: 0, 1, 2, 3, 4, 5...

Binary: 0, 1, 10, 11, 100, 101...



A cartoon illustration of a young boy with dark skin and hair, wearing an orange shirt and green pants. He has a lightbulb idea above his head, indicated by a yellow lightbulb with rays. A yellow thought bubble is also visible in the top left corner.

WHAT IS BINARY?



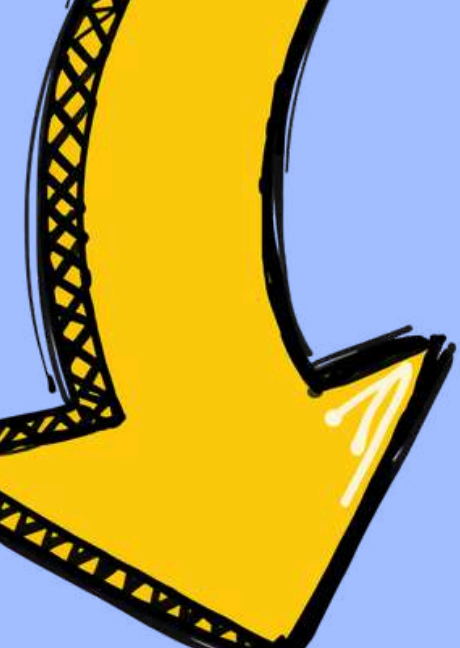
Binary: The Language of 0s and 1s

Binary is a number system that uses only two digits: 0 and 1. Every number, letter, and image in a computer is made from combinations of these two simple digits!

Why Computers Love Binary

Computers use binary because they work like light switches — either ON (1) or OFF (0). This simple on/off system makes computers fast and reliable!





HOW BINARY WORKS

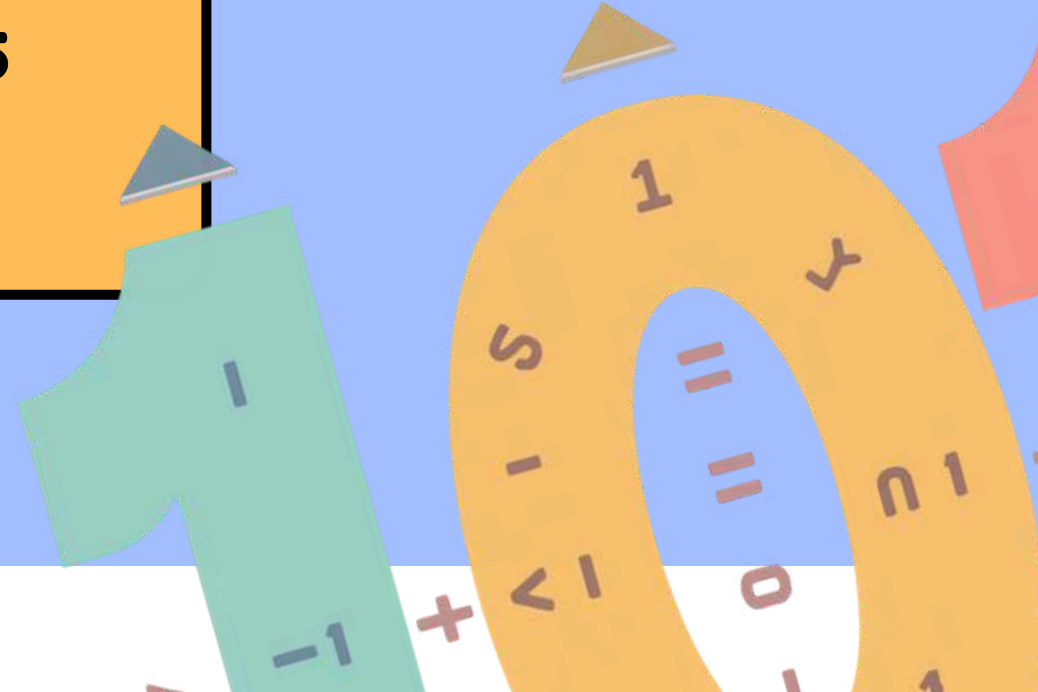


Each digit is called a BIT

Bits represent powers of
2

1, 2, 4, 8, 16, 32...

Example: Binary 101

$$1 \times 4 + 0 \times 2 + 1 \times 1 = 5$$


COUNTING IN BINARY

0 to 3 in Binary

0=0000, 1=0001, 2=0010, 3=0011

8 to 11 in Binary

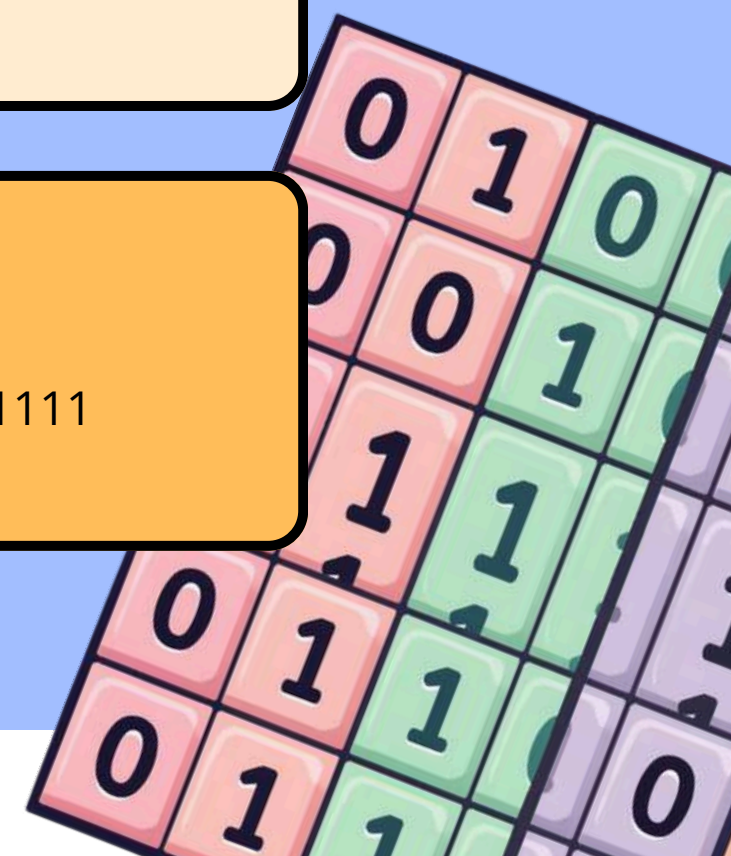
8=1000, 9=1001, 10=1010, 11=1011

4 to 7 in Binary

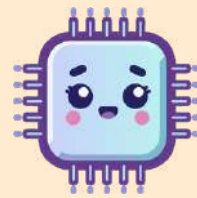
4=0100, 5=0101, 6=0110, 7=0111

12 to 15 in Binary

12=1100, 13=1101, 14=1110, 15=1111



WHY DO COMPUTERS USE BINARY?

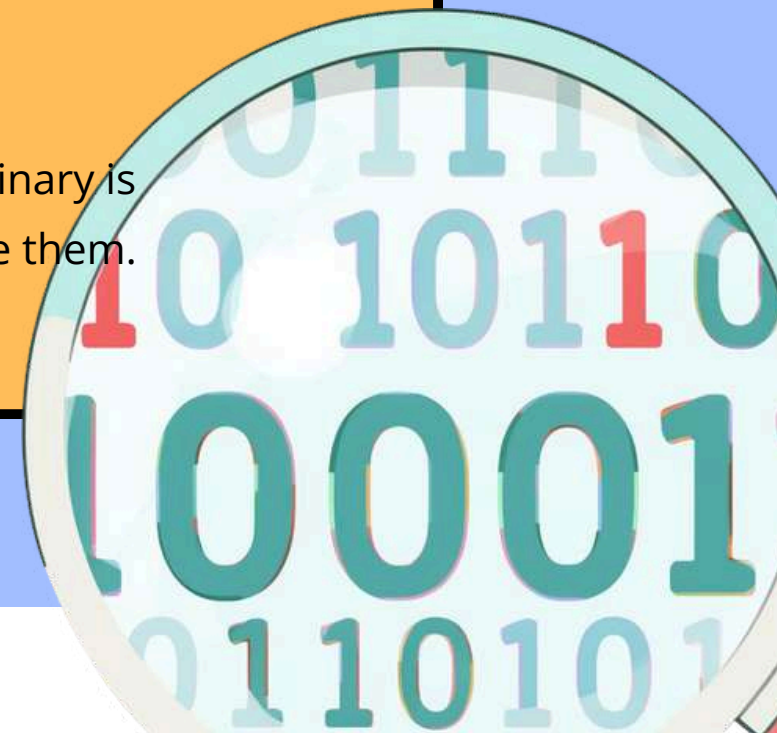
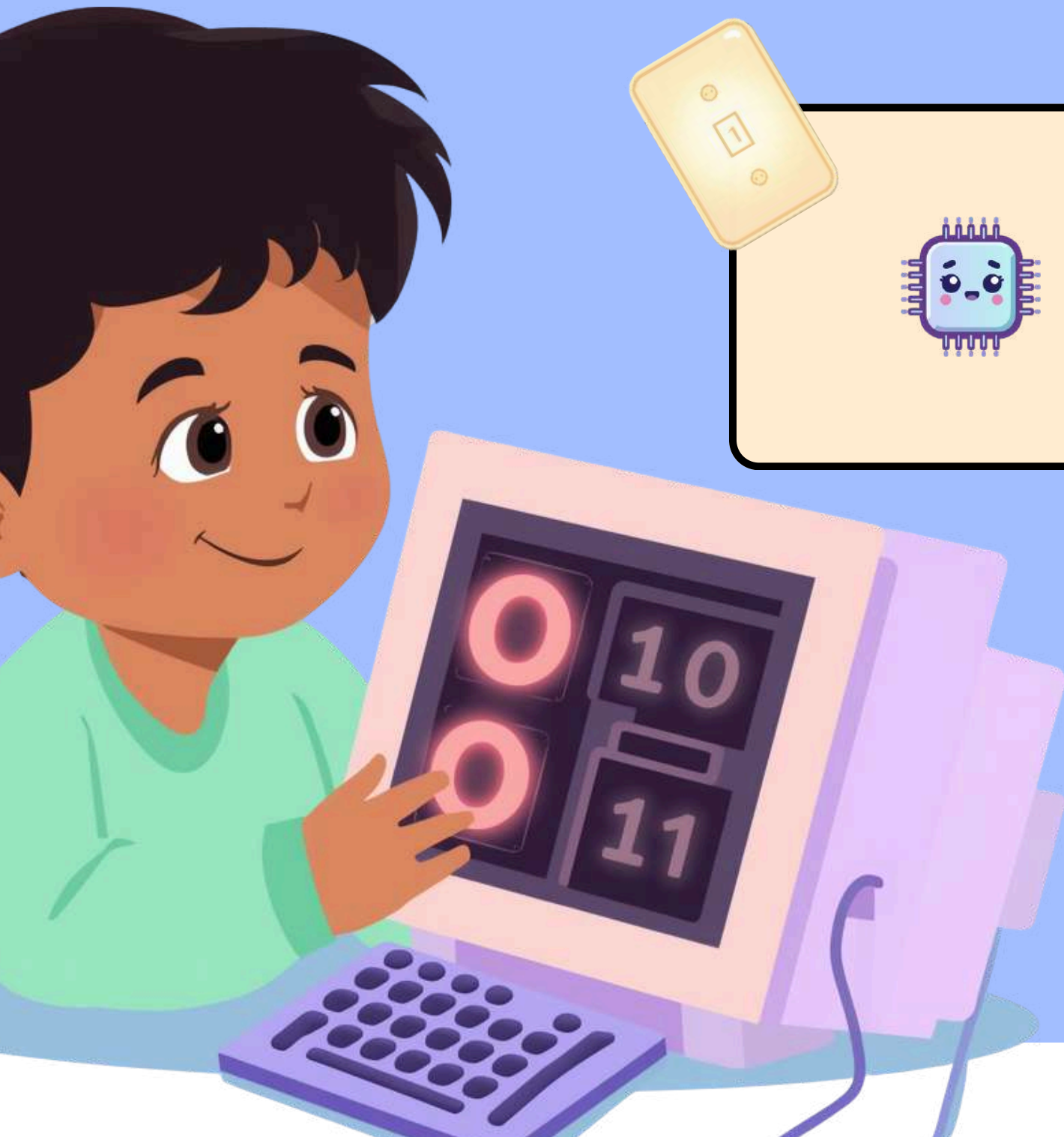


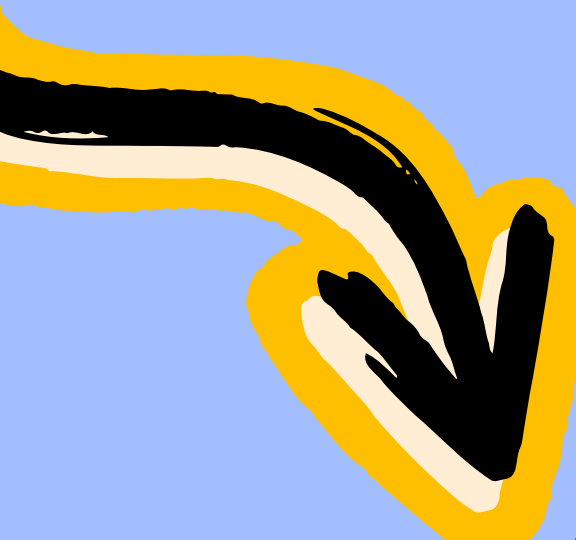
Simple and Reliable for Electronic Circuits

Electronic circuits work like tiny switches that can only be ON or OFF. Binary's two digits (0 and 1) match perfectly with these switches, making computers fast and dependable!

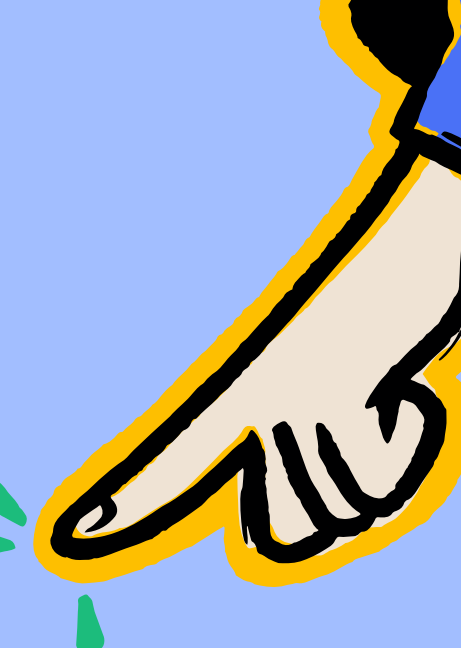
Easy to Store and Process Information

With only two options to remember, computers make fewer mistakes! Binary is less error-prone, helping your devices work smoothly every time you use them.





CONVERTING DECIMAL TO BINARY



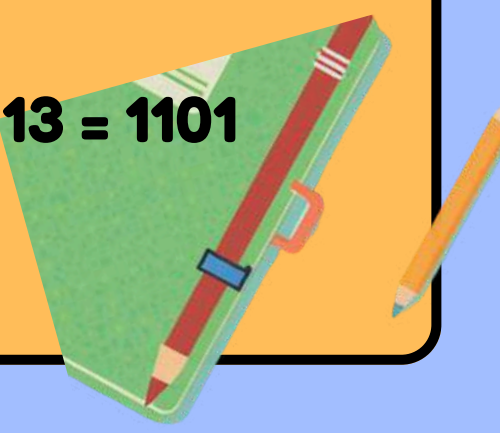
**Step 1: Divide the decimal
number by 2**

**Step 2: Write down the
remainder (0 or 1)**

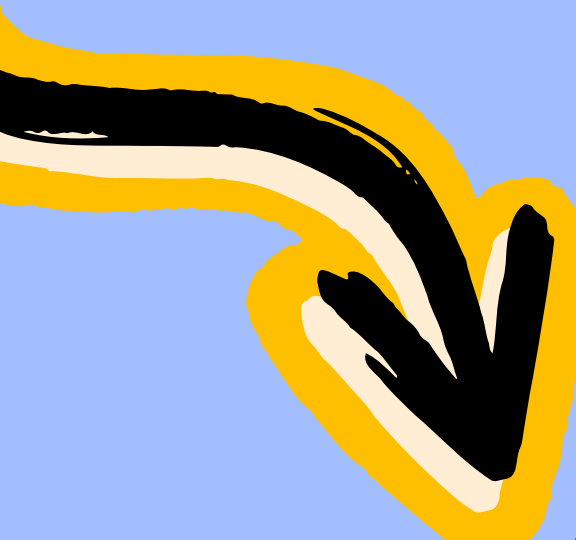
**Step 3: Repeat with
quotient until you reach 0**

**Step 4: Read remainders
backwards for binary**

Example: $13 = 1101$



13 → = 11



CONVERTING BINARY TO DECIMAL



Step 1: Write Powers of 2

Step 2: Multiply Each Bit

Step 3: Add All Results

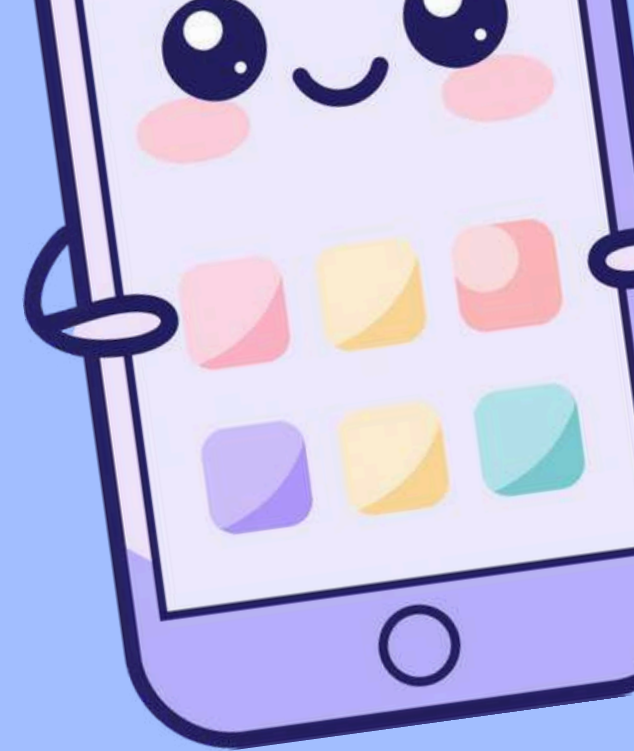
2

Example: 1010 in Binary

$$1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 = 10$$

BINARY to DE

1010 =



BINARY IN ACTION: AI & DIGITAL DEVICES



Voice Assistants

Binary helps AI assistants understand your voice and respond to questions!

Games & Apps

Every game and app uses binary code to create graphics and sounds

Smartphones

Your phone uses billions of binary bits to store photos, messages, and more!





FUN BINARY FACTS

**Computers use millions
of bits every second!**

**A byte equals 8 bits of
data**

**Binary creates images,
sounds & text**

**Binary is the language
of all digital devices**





Quiz Time!

TEST YOUR BINARY KNOWLEDGE!

Q1: What two digits are used in binary?

Q2: What decimal number is binary 1001?

Think carefully about what you learned today. Raise your hand when you know the answer!



RECAP: WHAT WE LEARNED

Binary Uses 0 and 1

Just two digits create all digital info!

Computers Count Binary

Machines store and process using bits

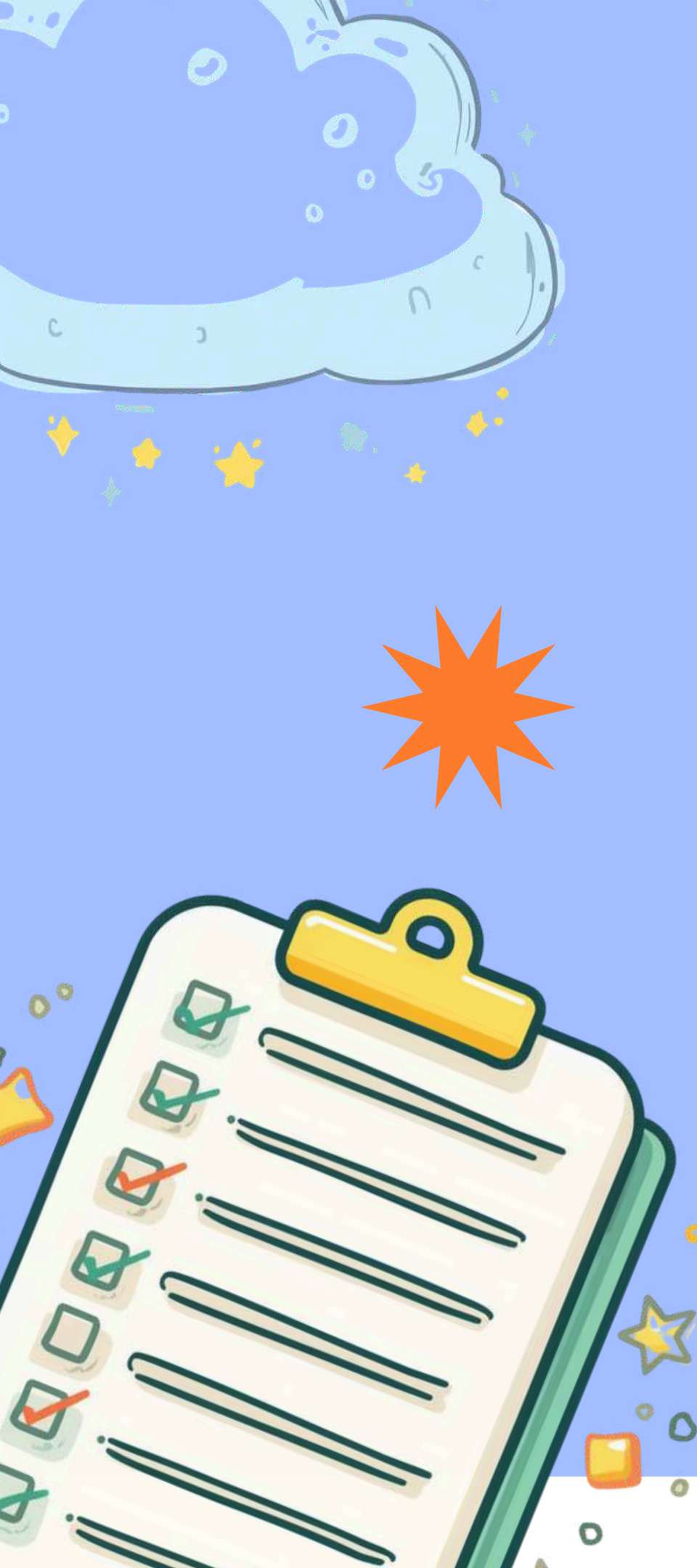
Easy Conversions

Decimal to binary and back again

Binary Is Everywhere

Phones, games, AI — all use binary!





KEEP EXPLORING!

Your Binary Journey Continues

Try counting in binary at home using your fingers!
Explore fun coding games online. Ask questions
about AI and computers — curiosity is the best
way to learn!





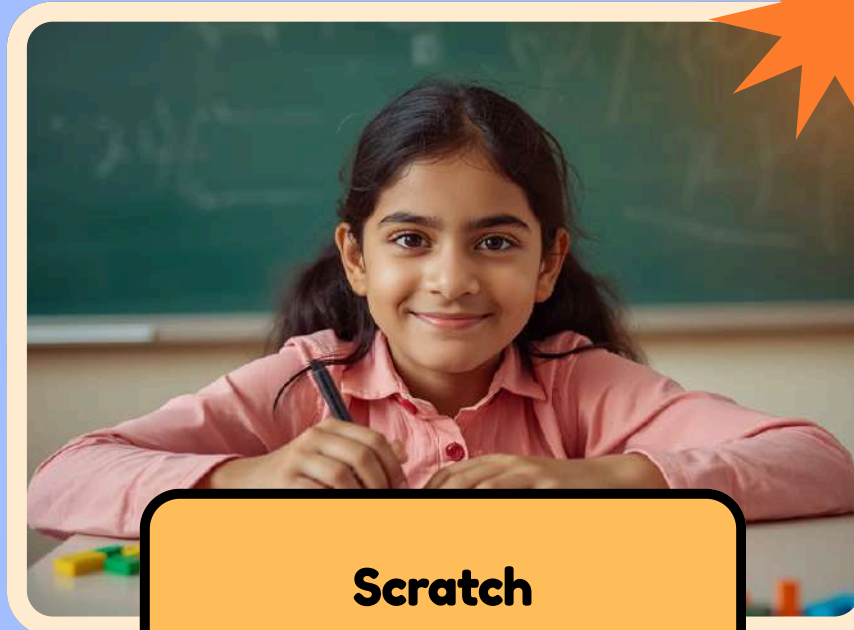
THANK YOU!

QUESTIONS? LET'S DISCUSS!

CREDITS AND RESOURCES



Code.org
Free Coding Games



Scratch
Kids Programming



Khan Academy
Binary Lessons